AGRIAI INTELLIGENCE SYSTEMS | CROP PROTECTION PLATFORM

End User & Farmer Guide

Operational Manual for Crop Diagnostics & IPM Advisory

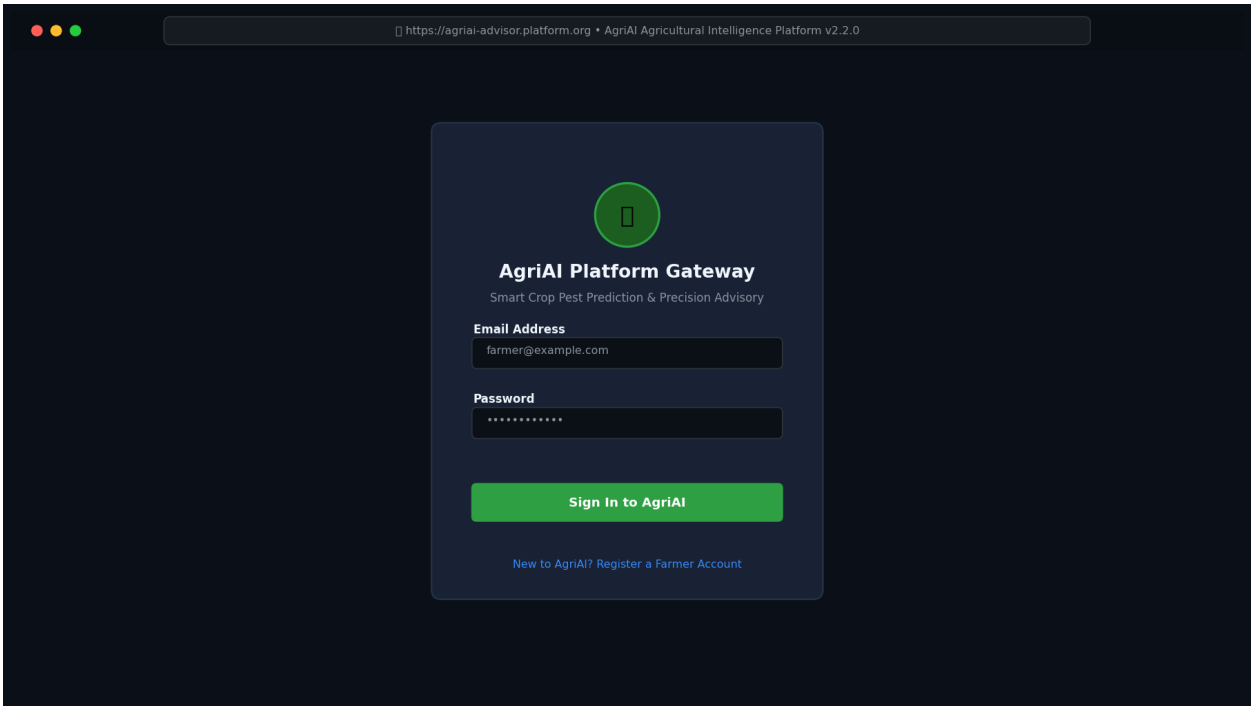
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Executive Summary: This document provides official specifications, standard operating procedures, and architectural blueprints for the AI-Powered Crop Pest Prediction & Agricultural Advisory System. The platform bridges machine learning multi-class diagnostics, agro-meteorological streaming data, and agronomic Integrated Pest Management (IPM) to protect farm yields sustainably.

1. System Access & Authentication

The AgriAI platform can be accessed from any desktop or mobile web browser. All user sessions are secured using encrypted password hashing and session state tokens.

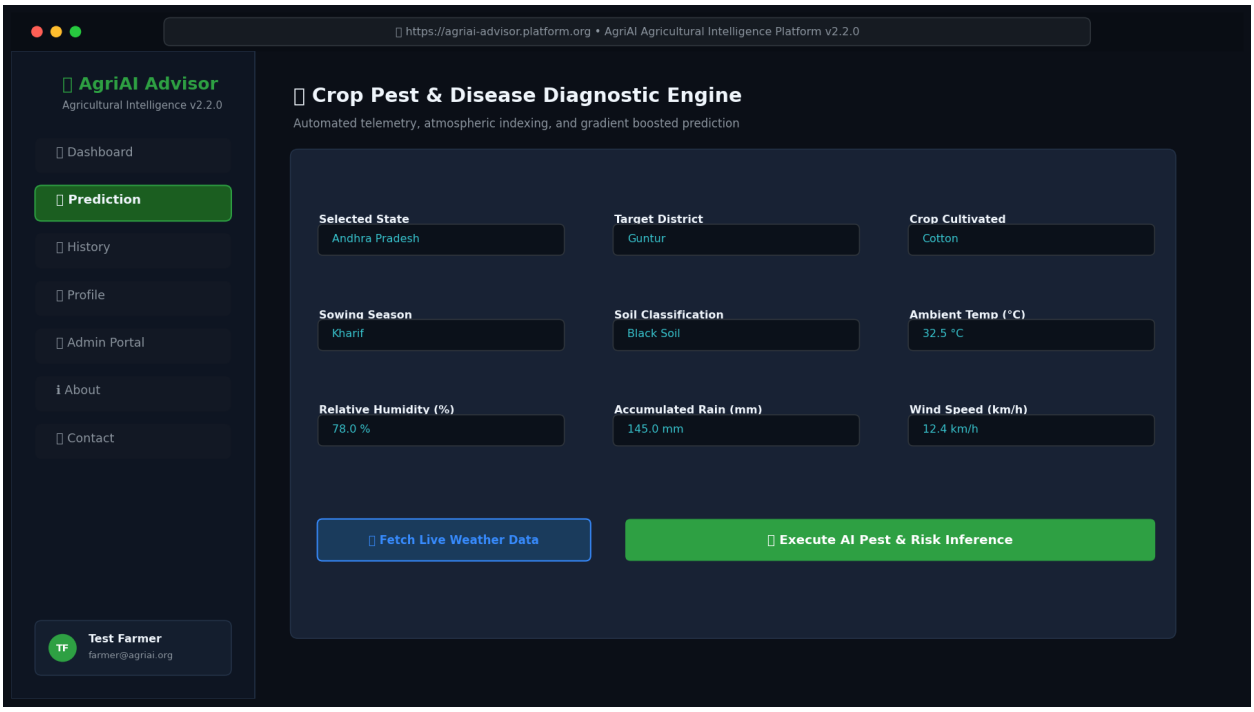
- **Account Registration:** Click on the Register tab in the left sidebar. Provide your Full Name, Email, Farm Location, and Primary Crop.
- **Secure Login:** Enter your registered email and password to access the farmer dashboard.
- **Role Privileges:** Farmers enjoy unlimited predictions, history tracking, and PDF downloads.



2. Pest Prediction Engine

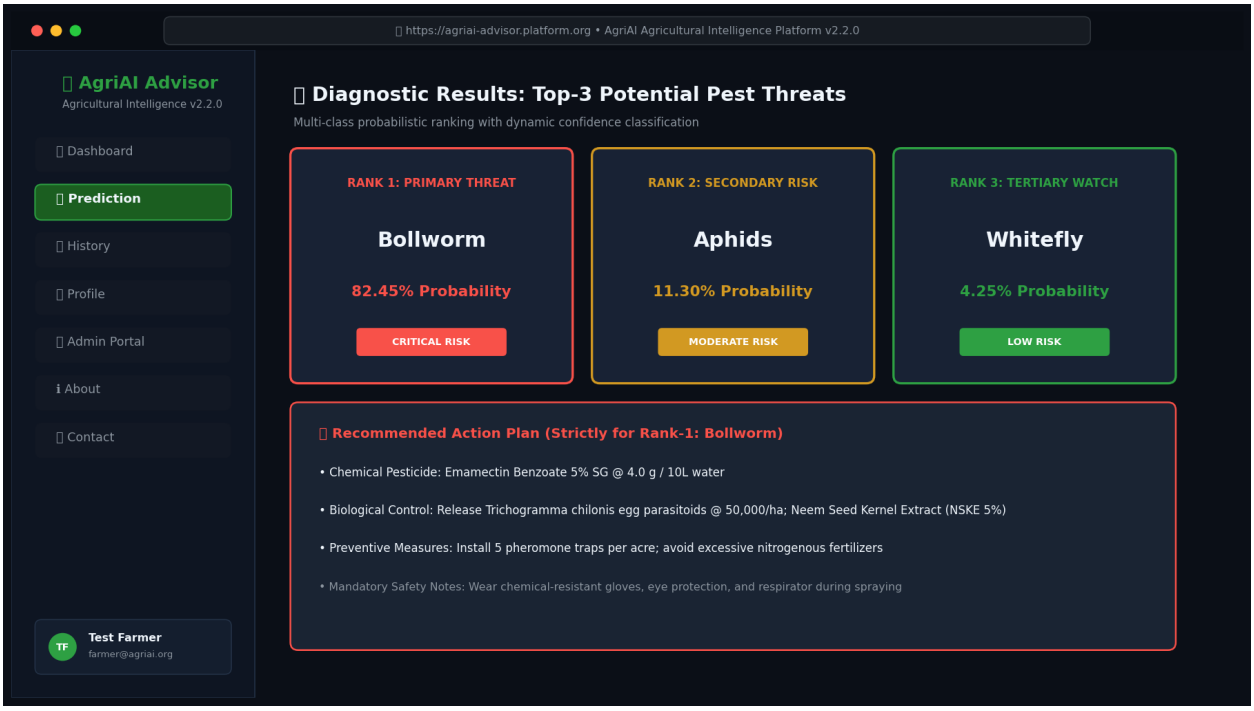
The core diagnostic engine accepts field parameters across three categories:

- **Crop Characteristics:** Select Crop Type (Wheat, Rice, Maize, etc.), Growth Stage, and Soil Type.
- **Atmospheric Conditions:** Temperature, Humidity, Rainfall, and Wind Speed can be entered manually or auto-populated using the **'Auto-Fetch Live Weather'** button.
- **Agronomic Context:** Input N-P-K nutrient status, irrigation method, and past infestation records.


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Test Farmer
farmer@agriai.org
Crop Pest & Disease Diagnostic Engine
Automated telemetry, atmospheric indexing, and gradient boosted prediction
Selected State
Andhra Pradesh
Target District
Guntur
Crop Cultivated
Cotton
Sowing Season
Kharif
Soil Classification
Black Soil
Ambient Temp (°C)
32.5 °C
Relative Humidity (%)
78.0 %
Accumulated Rain (mm)
145.0 mm
Wind Speed (km/h)
12.4 km/h
Fetch Live Weather Data
Execute AI Pest & Risk Inference

3. Interpreting Top-3 Predictions & Differential Diagnosis

Rather than returning a single black-box output, AgriAI returns a ranked Top-3 probability distribution. This enables agronomists to identify co-infestations or secondary threats early.



4. Generating Formal PDF Reports & Farm Audits

Every diagnostic prediction can be exported as a publication-ready Agronomic PDF Advisory Report. The report includes tamper-evident cryptographic hashes, timestamp, input metrics, Top-3 probability charts, and precise chemical dosage calculations.


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About
Contact
Test Farmer
farmer@agriai.org
Automated Agricultural Advisory Report (A4 PDF Generated)
Formal decision-support document exported via ReportLab vector rendering engine
AI CROP PEST ADVISORY REPORT
Farmer Name: Test Farmer | Date: 2026-10-02 | State: Andhra Pradesh | District: Guntur
1. PRIMARY DIAGNOSIS (RANK-1 THREAT)
Predicted Pest: Bollworm (Helicoverpa armigera) • Confidence: 82.45% (High Risk)
2. RECOMMENDED CHEMICAL CONTROLS & DOSAGE
• Apply Emamectin Benzoate 5% SG @ 4.0 g per 10 Liters water via knapsack sprayer.
• Apply Flubendiamide 39.35% SC @ 2.5 ml per 10 Liters water at egg-hatching phase.
3. ORGANIC & BIOLOGICAL CONTROL PROTOCOLS
• Release Trichogramma chilonis egg parasitoids @ 50,000 / hectare at weekly intervals.
• Spray Neem Seed Kernel Extract (NSKE 5%) during early larval instars.
4. MANDATORY FARMER SAFETY & PPE GUIDELINES
• Always wear chemical-resistant nitrile gloves, eye goggles, and a particulate respirator.
• Avoid spraying against prevailing wind direction; observe a 14-day pre-harvest interval.
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